

# MultiVariable Calculus

## Partial and Directional Derivatives

A multivariate function,  $f(x, y, z)$  exists in 3D space, so when looking for the gradient of  $f(x, y, z)$  we have to consider the slope when moving in a particular direction from any point  $(x, y, z)$  on  $f$ .

Partial derivatives allow us to consider the gradient when moving in the direction of one of the coordinate axes,  $x, y$  or  $z$ . To find the partial derivative wrt  $x, y$  or  $z$ , treat the other variables as unknown fixed constants and differentiate as normal.

Partial derivatives wrt  $x, y$  and  $z$  are defined as:

$$\frac{\partial f}{\partial x} = f_x = \lim_{h \rightarrow 0} \frac{f(x+h, y, z) - f(x, y, z)}{h}$$

$$\frac{\partial f}{\partial y} = f_y = \lim_{h \rightarrow 0} \frac{f(x, y+h, z) - f(x, y, z)}{h} \quad \text{ew, limits "}$$

$$\frac{\partial f}{\partial z} = f_z = \lim_{h \rightarrow 0} \frac{f(x, y, z+h) - f(x, y, z)}{h}$$

Standard differentiation rules apply with partial derivatives

### Example

Find all first-order partial derivatives of  $f(x, y, z) = \frac{x \sin y}{z^2}$

$$f_x = \frac{\partial}{\partial x} \left( \frac{x \sin y}{z^2} \right) = \frac{\partial}{\partial x} \left( \underbrace{\frac{\sin y}{z^2}}_{\text{constant}} x \right) = \frac{\sin y}{z^2}$$

$$f_y = \frac{\partial}{\partial y} \left( \underbrace{\frac{x}{z^2}}_{\text{const.}} \sin y \right) = \frac{x \cos y}{z^2} \quad \text{from standard differentiation: } \frac{d}{dy} \sin y = \cos y$$

$$f_z = \frac{\partial}{\partial z} \left( x \sin y z^{-2} \right) = -2x \sin y z^{-3}$$

Partial derivatives are actually a specific case of directional derivatives, which allow us to work out the derivative of  $f(x, y, z)$  in the direction of any unit vector  $\underline{v} = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix}$ . This is defined as:

$$\text{Directional derivative of } f(x, y, z) \text{ in direction } \underline{v} = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix}, \quad \nabla_{\underline{v}} f(x, y, z) = \lim_{h \rightarrow 0} \frac{f(x+h \cdot v_x, y+h \cdot v_y, z+h \cdot v_z) - f(x, y, z)}{h}$$

Or:

$$\nabla_{\underline{v}} f(x, y, z) = v_x f_x + v_y f_y + v_z f_z = \underline{v} \cdot \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix}$$

Note:  $\underline{v}$  has to be a unit vector

This vector is called  $\nabla f$ .

$\nabla f$  is a vector in the direction of greatest rate of increase of  $f$ , in terms of  $x, y, z$ .

The magnitude of  $\nabla f(x_0, y_0, z_0)$  at a point  $(x_0, y_0, z_0)$  is this greatest rate of increase at that point, with the value of  $\nabla f$  being the direction.

Note: Partial and directional derivatives work the same way for any number of variables

### Example

- Find  $\nabla f$ , where  $f(w, x, y, z) = w^2 - 2x^3 + 3yz$
- Use this value of  $\nabla f$  to find the directional derivative of  $f$  at  $(-2, 0, -1, 1)$  in the direction  $\underline{v} = \begin{pmatrix} 2 \\ 6 \\ -4 \\ 5 \end{pmatrix}$
- Find the direction of greatest increase of  $f$  at  $(-2, 0, -1, 1)$

$$\begin{aligned} \textcircled{1} \quad & F_w = 2w \\ & F_x = -6x^2 \\ & F_y = 3z \\ & F_z = 3y \end{aligned} \quad \therefore \nabla f = \begin{pmatrix} F_w \\ F_x \\ F_y \\ F_z \end{pmatrix} = \begin{pmatrix} 2w \\ -6x^2 \\ 3z \\ 3y \end{pmatrix}$$

$$\textcircled{2} \quad \text{Directional derivative} = \underline{v} \cdot \nabla f$$

need  $\underline{v}$  to be a unit vector

$$\underline{v} = \begin{pmatrix} 2 \\ 6 \\ -4 \\ 5 \end{pmatrix}$$

$$\|\underline{v}\| = \sqrt{2^2 + 6^2 + (-4)^2 + 5^2} = \sqrt{81} = 9$$

$$\therefore \text{use } \underline{v} = \begin{pmatrix} 2/9 \\ 6/9 \\ -4/9 \\ 5/9 \end{pmatrix}$$

need to sub in  $(-2, 0, -1, 1)$  into  $\nabla f$ :

$$\nabla f(-2, 0, -1, 1) = \begin{pmatrix} -4 \\ 0 \\ 3 \\ -3 \end{pmatrix}$$

$$\text{Directional derivative} \quad \begin{pmatrix} 2/9 \\ 6/9 \\ -4/9 \\ 5/9 \end{pmatrix} \cdot \begin{pmatrix} -4 \\ 0 \\ 3 \\ -3 \end{pmatrix} = -8/9 - 12/9 - 15/9 = \boxed{-35/9}$$

!!

$$\textcircled{3} \quad \text{Direction of greatest increase} = \nabla f(-2, 0, -1, 1) = \begin{pmatrix} -4 \\ 0 \\ 3 \\ -3 \end{pmatrix}$$

## Second Order Partial Derivatives and Extrema

We can find the second order partial derivatives of a multivariate function  $f(x, y, z)$  by finding the partial derivatives of  $F_x, F_y$  and  $F_z$ :

$$F_{xx} = \frac{\partial}{\partial x}(F_x) \quad F_{yx} = \frac{\partial}{\partial y}(F_x) \quad F_{zx} = \frac{\partial}{\partial z}(F_x)$$

$$F_{xy} = \frac{\partial}{\partial x}(F_y) \quad \text{etc...}$$

This is the same for any  $F$  with any number of variables.

Note: the ppts have  $F_{yx} = \frac{\partial}{\partial y}(F_x)$  but google has  $F_{yx} = \frac{\partial}{\partial x}(F_y)$  so fuck knows who's right 😒

These second order partial derivatives can be gathered into the Hessian matrix,  $H_F$ :

$$H_F = \begin{pmatrix} \frac{\partial}{\partial x}(F_x) & \frac{\partial}{\partial x}(F_y) & \frac{\partial}{\partial x}(F_z) \\ \frac{\partial}{\partial y}(F_x) & \frac{\partial}{\partial y}(F_y) & \frac{\partial}{\partial y}(F_z) \\ \frac{\partial}{\partial z}(F_x) & \frac{\partial}{\partial z}(F_y) & \frac{\partial}{\partial z}(F_z) \end{pmatrix}$$

$$\text{Note: usually } \frac{\partial}{\partial x}(F_y) = \frac{\partial}{\partial y}(F_x)$$

$H_F$  can be used to determine the nature of stationary points (extrema) of  $f$ :

Extrema of  $f$  can be found by finding all points  $(x_0, y_0, z_0)$  such that  $f_x = f_y = f_z = 0$

To determine whether each stationary point  $(x_0, y_0, z_0)$  is a local minimum, a local maximum or a saddle point, substitute  $x = x_0, y = y_0, z = z_0$  into  $H_f$  and find the eigenvalues of  $H_f$  with each point subbed in.

—  $(x_0, y_0, z_0)$  is a local minimum if all eigenvalues of  $H_f(x_0, y_0, z_0)$  are positive

—  $(x_0, y_0, z_0)$  is a local maximum if all eigenvalues of  $H_f(x_0, y_0, z_0)$  are negative

—  $(x_0, y_0, z_0)$  is a saddle point if  $H_f(x_0, y_0, z_0)$  has positive and negative eigenvalues

Note: If any eigenvalues are zero, you're fucked (and cannot determine anything about the point).

^ This applies to functions with any number of variables

Shortcut for  $g(x, y)$  with two variables:

Substitute  $x = x_0, y = y_0$  into  $H_g$  (where  $(x_0, y_0)$  is a stationary point) and calculate  $\det(H_g)$ .

—  $(x_0, y_0)$  is a local minimum if  $\det(H_g) > 0$  and  $f_{xx} < 0$  at  $(x_0, y_0)$

—  $(x_0, y_0)$  is a local maximum if  $\det(H_g) > 0$  and  $f_{xx} > 0$  at  $(x_0, y_0)$

—  $(x_0, y_0)$  is a saddle point if  $\det(H_g) < 0$  at  $(x_0, y_0)$

Notes: - if  $\det(H_g) = 0$  you're fucked

$$-\det(H_g) = f_{xx}f_{yy} - f_{xy}f_{yx} = f_{xx}f_{yy} - f_{xy}^2 \text{ as } f_{ky} = f_{yk} \text{ in most cases}$$

### Example (3 variables)

Identify the nature of the stationary points of  $F(x, y, z) = x^2 + y^2 + z^2 - xy - z$

Find first-order partial derivatives:

$$f_x = 2x - y$$

$$f_y = 2y - x$$

$$f_z = 2z - 1$$

Set  $f_x, f_y, f_z$  equal to zero to find stationary points:

$$2x - y = 0 \quad 2z - 1 = 0$$

$$2y - x = 0 \quad z = 1$$

$$-2y + 4x = 0$$

$$3x = 0$$

$$x = 0$$

$$y = 0$$

$\therefore$  one stationary point  $x_0 = (0, 0, 1)$

Find second order partial derivatives and  $H_f$ .

$$f_{xx} = 2 \quad f_{xy} = -1 \quad f_{xz} = 0$$

$$f_{yx} = -1 \quad f_{yy} = 2 \quad f_{yz} = 0$$

$$f_{zx} = 0 \quad f_{zy} = 0 \quad f_{zz} = 2$$

$$\therefore H_f = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

No need to sub in  $\underline{x}_0$  as all terms in  $H_f$  constant in this case

Find eigenvalues of  $H_f$ :

$$\det(H_f - \lambda I) = \det \begin{pmatrix} 2-\lambda & -1 & 0 \\ -1 & 2-\lambda & 0 \\ 0 & 0 & 2-\lambda \end{pmatrix} = 0$$

$$(2-\lambda)((2-\lambda)^2 - 1) = (2-\lambda)^3 - (2-\lambda) = 0$$

$$(2-\lambda)((2-\lambda)^2 - 1) = 0$$

$$(2-\lambda)(\lambda^2 - 4\lambda + 3) = 0$$

$$(2-\lambda)(\lambda-3)(\lambda-1) = 0$$

$$\lambda = 2, \lambda = 3, \lambda = 1 \rightarrow \text{all positive}$$

$\therefore$  Stationary point  $\underline{x}_0 = (0, 0, 1)$  is a local minimum.

We can use the partial derivatives and the Hessian matrix to find linear and quadratic approximations for a multivariate function  $f(x, y, z)$  at a point  $\underline{x}_0 = (a, b, c)$

- The general linear approximation of  $f$  at  $\underline{x}_0$  is given by:

$$f(x, y, z) \approx f(\underline{x}_0) + \nabla f(\underline{x}_0)(\underline{x} - \underline{x}_0) \rightarrow f_x(x, y, z)(x-a) + f_y(x, y, z)(y-b) + f_z(x, y, z)(z-c)$$

- The general quadratic approximation of  $f$  at  $\underline{x}_0$  is given by:

$$f(x, y, z) \approx f(\underline{x}_0) + \nabla f(\underline{x}_0)(\underline{x} - \underline{x}_0) + \frac{1}{2}(\underline{x} - \underline{x}_0)^T H_f(\underline{x}_0)(\underline{x} - \underline{x}_0)$$

Matrix multiplication

- If  $\underline{x}_0$  is a stationary point,  $\nabla f(\underline{x}_0) = 0$ , so the quadratic approximation of  $f$  at a stationary point  $\underline{x}_0$  is:

$$f(x, y, z) \approx f(\underline{x}_0) + \frac{1}{2}(\underline{x} - \underline{x}_0)^T H_f(\underline{x}_0)(\underline{x} - \underline{x}_0)$$

$$\text{Note: } (\underline{x} - \underline{x}_0) = \begin{pmatrix} x-a \\ y-b \\ z-c \end{pmatrix}, (\underline{x} - \underline{x}_0)^T = (x-a \ y-b \ z-c)$$

Example:

$$\text{Consider } f(x, y) = \frac{1}{2} \sin(\pi x) - \cos(2\pi y)$$

① Determine a function  $g(x, y)$  that gives the best linear approximation of  $f$  about the point  $(\frac{1}{2}, 2)$

② Determine a function  $h(x, y)$  that gives the best quadratic approximation of  $f$  about the point  $(\frac{1}{2}, 1)$

① Find  $\nabla f$ :

$$f_x = \frac{\pi}{2} \cos(\pi x)$$

$$f_y = 2\pi \sin(2\pi y)$$

$$\nabla f = \begin{pmatrix} \frac{\pi}{2} \cos(\pi x) \\ 2\pi \sin(2\pi y) \end{pmatrix}$$



Get  $g(x, y)$

$$\begin{aligned} g(x, y) &= F(x_0) + \nabla F(x_0)(x - x_0) \\ &= F(1, 2) + f_x(1, 2)(x - 1) + f_y(1, 2)(y - 2) \\ &= \frac{1}{2} \sin \pi - \cos(4\pi) + \frac{\pi}{2} \cos \pi (x - 1) + 2\pi \sin(4\pi)(y - 2) \\ g(x, y) &= -1 - \frac{\pi}{2}(x - 1) \end{aligned}$$

(2) Find  $\nabla F$  at  $(\frac{1}{2}, 1)$ :

$$\begin{aligned} f_x(\frac{1}{2}, 1) &= \frac{\pi}{2} \cos \pi \frac{1}{2} = 0 \rightarrow (\frac{1}{2}, 1) \text{ is a stationary point !!} \\ f_y(\frac{1}{2}, 1) &= 2\pi \sin 2\pi = 0 \end{aligned}$$

Find  $H_F$ :

$$f_x = \frac{\pi}{2} \cos(\pi x), \quad f_y = 2\pi \sin(2\pi y)$$

$$f_{xx} = -\frac{\pi^2}{2} \sin(\pi x), \quad f_{xy} = 0$$

$$f_{yx} = 0, \quad f_{yy} = 4\pi^2 \cos(2\pi y)$$

$$H_F = \begin{pmatrix} -\frac{\pi^2}{2} \sin(\pi x) & 0 \\ 0 & 4\pi^2 \cos(2\pi y) \end{pmatrix}$$

At  $(\frac{1}{2}, 1)$ :

$$H_F(\frac{1}{2}, 1) = \begin{pmatrix} -\frac{\pi^2}{2} & 0 \\ 0 & 4\pi^2 \end{pmatrix}$$

Get  $h(x, y)$  at  $x_0 = (\frac{1}{2}, 1)$

$$h(x, y) = F(x_0) + \frac{1}{2}(x - x_0)^T H_F(x_0)(x - x_0) \rightarrow (\frac{1}{2}, 1) \text{ is stationary so the linear term goes to zero, so can be ignored !!}$$

$$= \frac{1}{2} \sin(\frac{\pi}{2}) - \cos(2\pi) + \frac{1}{2} \begin{pmatrix} x - \frac{1}{2} & y - 1 \end{pmatrix} \begin{pmatrix} -\frac{\pi^2}{2} & 0 \\ 0 & 4\pi^2 \end{pmatrix} \begin{pmatrix} x - \frac{1}{2} \\ y - 1 \end{pmatrix}$$

$$= \frac{1}{2} - 1 + \frac{1}{2} \begin{pmatrix} -\frac{\pi^2}{2}(x - \frac{1}{2}) & 4\pi^2(y - 1) \end{pmatrix} \begin{pmatrix} x - \frac{1}{2} \\ y - 1 \end{pmatrix}$$

$$= -\frac{1}{2} + \frac{1}{2} \left( -\frac{\pi^2}{2}(x - \frac{1}{2})^2 + 4\pi^2(y - 1)^2 \right)$$

$$h(x, y) = -\frac{1}{2} - \frac{\pi^2}{4}(x - \frac{1}{2})^2 + 2\pi^2(y - 1)^2$$

## Vector Valued Functions

So far, we have considered functions of the form  $F(x_1, \dots, x_n) = a \in \mathbb{R}$ , which outputs a single real number  $a$  determined by the values of the inputs  $x_1, \dots, x_n$ . These inputs can be thought of as a single input  $n$ -dimensional vector, and as such,  $F$  can be thought of as a mapping from  $\mathbb{R}^n$  to  $\mathbb{R}$ , i.e.:

$$F: \mathbb{R}^n \rightarrow \mathbb{R}$$

A **vector valued function** is one that maps from  $\mathbb{R}^n$  to  $\mathbb{R}^m$ , i.e. a vector valued function  $g(x_1, \dots, x_n) = \begin{pmatrix} f_1 \\ \vdots \\ f_m \end{pmatrix}$ ,  $g: \mathbb{R}^n \rightarrow \mathbb{R}^m$ .

For example:  $F(x, y) = (x^2y - z, x - y^3)$

$$g(x) = (x^2, x + 1)$$

$$h(w, x, y, z) = (wx + yz, z^2 - 3w, (y + 2x)^2)$$

$$\begin{aligned} f: \mathbb{R}^2 &\rightarrow \mathbb{R}^2 \\ g: \mathbb{R} &\rightarrow \mathbb{R}^2 \\ h: \mathbb{R}^4 &\rightarrow \mathbb{R}^3 \end{aligned}$$

( $n$  is the number of variables,  $m$  is the number of equations)

For a vector valued function  $F(x_1, \dots, x_n) = (f_1(x_1, \dots, x_n), \dots, f_m(x_1, \dots, x_n))$ , the partial derivative of  $F$  with respect to  $x_i$  ( $1 \leq i \leq n$ ) can be thought of as the  $m$ -dimensional vector, with the  $j$ <sup>th</sup> entry containing the partial derivative of  $f_j$  wrt  $x_i$ , i.e.:

$$\frac{\partial F}{\partial x_i} = \begin{pmatrix} \frac{\partial f_1}{\partial x_i} \\ \vdots \\ \frac{\partial f_m}{\partial x_i} \end{pmatrix}$$

If we do this with all of  $x_1, \dots, x_n$ , these vectors can be collected as columns of a matrix: the **Jacobian**,  $J_F$ :

$$J_F = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \dots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}$$

Notes: - The  $i$ <sup>th</sup> row of  $J_F$  is the gradient,  $\nabla f_i$ , of  $f_i$   
 -  $J_F$  isn't always square - Very often  $m \neq n$ .

This **Jacobian matrix** can be used to find the best **linear approximation** of a multivariate vector-valued function at a point.

The linear approximation of a function  $F(\underline{x})$ ,  $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$  near a point  $\underline{x}_0 = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$  is defined as:

$$F(\underline{x}) \approx F(\underline{x}_0) + J_F(\underline{x}_0)(\underline{x} - \underline{x}_0)$$

Where  $\underline{x} - \underline{x}_0 = \begin{pmatrix} x_1 - a_1 \\ \vdots \\ x_n - a_n \end{pmatrix}$ ,  $J_F(\underline{x}_0)$  is the **Jacobian matrix** with  $\underline{x}_0$  substituted in.

### Example

Find the best linear approximation of  $F(x, y) = \begin{pmatrix} x^2 + y \\ xy \end{pmatrix}$  near the point  $(1, 2)$

$\begin{matrix} \parallel & \parallel \\ f_1 & f_2 \end{matrix}$

Compute  $J_F$ :

$$\frac{\partial f_1}{\partial x} = 2x \quad \frac{\partial f_1}{\partial y} = 1$$

$$\frac{\partial f_2}{\partial x} = y \quad \frac{\partial f_2}{\partial y} = x$$

$$\therefore J_F = \begin{pmatrix} 2x & 1 \\ y & x \end{pmatrix}$$

Evaluate linear approximation formula at  $(1, 2)$ :

$$F(x, y) \approx F(1, 2) + J_F(1, 2) \begin{pmatrix} x - 1 \\ y - 2 \end{pmatrix}$$

$$F(x, y) \approx \begin{pmatrix} 3 \\ 2 \end{pmatrix} + \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x - 1 \\ y - 2 \end{pmatrix}$$

$$F(x, y) \approx \begin{pmatrix} 3 + 2(x - 1) + (y - 2) \\ 2 + 2(x - 1) + (y - 2) \end{pmatrix}$$

# Lagrangian Multipliers

If we want to maximise or minimise a multivariate function  $f(x_1, \dots, x_n)$ , with variables  $x_1, \dots, x_n$  independent, we would consider the stationary points of  $f$  and look for maxima or minima. However, if we subject  $x_1, \dots, x_n$  to a constraint,  $g(x_1, \dots, x_n)$ , i.e.  $x_1, \dots, x_n$  are no longer independent, this will no longer work, as the minima/maxima of  $f$  are unlikely to satisfy  $g$ . We need a different method to find these extrema:

Let  $f(x_1, \dots, x_n) = k$  be subject to a constraint,  $g(x_1, \dots, x_n) = 0$ . Geometrically, points on  $f$  that satisfy  $g$  are those that lie on the intersection of the graphs of  $f(x_1, \dots, x_n) = k$  and  $g(x_1, \dots, x_n) = 0$ . To find extrema in this case, we need to find values of  $k$  where  $f$  just touches  $g$ . These values of  $k$  are the minimum/maximum values of  $f$  subject to  $g$ . These points where  $f(x_1, \dots, x_n)$  just touch are our constrained extrema.

At these points where  $f(x_1, \dots, x_n) = k$  just touches  $g(x_1, \dots, x_n)$ ,  $f$  and  $g$  are tangential, i.e. the gradient vectors  $\nabla f$  and  $\nabla g$  are parallel, so:

$$\nabla f = \lambda \nabla g \text{ for some } \lambda \in \mathbb{R}$$

From this we can obtain a system of equations:

$$\left. \begin{array}{l} f_{x_1} = \lambda g_{x_1} \\ \vdots \\ f_{x_n} = \lambda g_{x_n} \\ g = 0 \end{array} \right\} \begin{array}{l} n+1 \text{ equations} \\ n+1 \text{ variables } (\lambda, x_1, \dots, x_n) \end{array}$$

Which we can solve for  $x_1, \dots, x_n$ , which are our constrained maxima/minima of  $f$  subject to  $g$ .  
Equivalently:

You can max/min-imise a multivariate function  $f(x_1, \dots, x_n)$  subject to a constraint  $g(x_1, \dots, x_n) = 0$  by defining the Lagrangian:

$$L(x_1, \dots, x_n, \lambda) = f(x_1, \dots, x_n) - \lambda g(x_1, \dots, x_n)$$

and finding its stationary points wrt  $x_1, \dots, x_n$  and  $\lambda$ , i.e. solving the system of equations:

$$\frac{\partial L}{\partial x_1} = 0$$

$$\vdots$$

$$\frac{\partial L}{\partial x_n} = 0$$

$$\frac{\partial L}{\partial \lambda} = 0 \rightarrow \text{equivalent to } g(x_1, \dots, x_n) = 0 \text{ since } \frac{\partial L}{\partial \lambda} = -g(x_1, \dots, x_n) = 0$$

Then, identify the nature of these constrained extrema, i.e. whether they are maxima or minima.

## Example 1

Maximise  $f(x, y, z) = xyz$  subject to  $x + z = 1 - y$

Get constraint in the form  $g(x, y, z) = 0$ :

$$g(x, y, z) = x + y + z - 1$$

Define Lagrangian:

$$L(x, y, z, \lambda) = xyz - \lambda(x + y + z - 1)$$

Take partial derivatives:

$$\frac{\partial L}{\partial x} = yz - \lambda = 0 \quad (1)$$

$$\frac{\partial L}{\partial y} = xz - \lambda = 0 \quad (2)$$

$$\frac{\partial L}{\partial z} = xy - \lambda = 0 \quad (3)$$

$$\frac{\partial L}{\partial \lambda} = -(x+y+z-1) = 1+y+z-1=0 \quad (4)$$

Solve:

$$\left. \begin{array}{l} \lambda = yz \\ \lambda = xz \end{array} \right\} \begin{array}{l} xz = yz \\ x = y \end{array} \quad \left. \begin{array}{l} \lambda = xz \\ \lambda = xy \end{array} \right\} \begin{array}{l} xz = xy \\ y = z \end{array} \quad \left. \begin{array}{l} x = y \\ y = z \end{array} \right\} x = y = z$$

$$x+y+z=1$$

$$3x = 1$$

$$x = \frac{1}{3}$$

$$y = \frac{1}{3}$$

$$z = \frac{1}{3}$$

$$\therefore \text{maximum at } \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

$$\text{maximum value of } f \text{ subject to } x+y+z=1: \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} = \frac{1}{27}$$

## Example 2

$$\text{Minimise } f(x,y) = x^2 \text{ subject to } x^2 + y^2 = 1$$

$$\text{Constraint } g(x,y) = 0:$$

$$g(x,y) = x^2 + y^2 - 1 = 0$$

Define Lagrangian:

$$L(x,y,\lambda) = x^2 - \lambda(x^2 + y^2 - 1)$$

Partial derivatives:

$$\frac{\partial L}{\partial x} = 2x - 2x\lambda = 0 \Rightarrow x(1-\lambda) = 0 \quad (1)$$

$$\frac{\partial L}{\partial y} = -2y\lambda = 0 \Rightarrow \lambda y = 0 \quad (2)$$

$$\frac{\partial L}{\partial \lambda} = -(x^2 + y^2 - 1) = 0 \Rightarrow x^2 + y^2 = 1 \quad (3)$$

Solve:

from (1):

$$x(1-\lambda) = 0$$

$$x = 0 \text{ or } \lambda = 1$$

From (2):

$$\lambda y = 0$$

$$\lambda = 0 \text{ or } y = 0$$

$$\text{If } \lambda = 1, y = 0, \text{ So from constraint } x^2 + y^2 = 1 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$$

$$\therefore x = \pm 1, y = 0$$

$$\text{In this case, } f(x,y) = (\pm 1)^2 = 1$$

$$\text{If } \lambda = 0, x = 0, \text{ So from constraint } x^2 + y^2 = 1 \Rightarrow y^2 = 1 \Rightarrow y = \pm 1$$

$$\therefore x = 0, y = \pm 1$$

$$\text{In this case, } f(x,y) = 0^2 = 0 < 1$$

$\therefore$  this is the minimum case

Not all solutions to the Lagrangian are minima, in this example both maxima and minima are present so be sure to evaluate  $f$  in each case to check which are maxima and minima, depending on which you want.

$\therefore$  Minima of  $F(x,y)$  subject to  $x^2+y^2=1$  occur at  $(0,-1)$  and  $(0,1)$   
minimum value of  $F(x,y)$  subject to  $x^2+y^2=1$  is 0